

Cloud Computing

Mock Exam 2 | Course ID: noc26_cs55

Q1. What does XaaS stand for in cloud computing?

- A) eXtra as a Service
- B) Anything as a Service
- C) XML as a Service
- D) eXecution as a Service

Source: Lecture_W1L19_Lecture_1__Cloud_Computing__Overview | Difficulty: Easy

Q2. Which of the following is NOT one of the five essential characteristics of cloud computing according to the NIST definition?

- A) On-Demand Self-Service
- B) Broad Network Access
- C) Resource Pooling
- D) Dedicated Hardware

Source: Lecture_W1L21_Lecture_3__Cloud_Computing__Introduction | Difficulty: Easy

Q3. Which cloud deployment model is provisioned for open use by the general public and operated by a third-party provider?

- A) Private Cloud
- B) Public Cloud
- C) Community Cloud
- D) Hybrid Cloud

Source: Lecture_W2L28_Lecture_6__Cloud_Computing__Architecture__Deploym | Difficulty: Easy

Q4. What is the primary purpose of XML (Extensible Markup Language) in cloud computing?

- A) To display data visually for human users.
- B) To structure and transport data for machine-to-machine communication.
- C) To execute programming logic on web servers.
- D) To manage network configurations.

Source: Lecture_W2L30_Lecture_8__Cloud_Computing__XML_Basics | Difficulty: Easy

Q5. Which XML-based language acts as the 'user manual' or contract for a web service, describing its capabilities and how to interact with it?

- A) SOAP
- B) XML
- C) WSDL
- D) UDDI

Source: Lecture_W2L32_Lecture_10__Cloud_Computing__Web_Services__Service | Difficulty: Easy

Q6. In OpenStack, which component is responsible for managing the entire lifecycle of virtual machines (VMs)?

- A) Glance
- B) Cinder
- C) Nova
- D) Neutron

Source: Lecture_W3L41_Lecture_15__Open_Stack | Difficulty: Easy

Q7. Which component of the CIA Triad ensures that data remains unaltered and accurate?

- A) Confidentiality
- B) Integrity
- C) Availability
- D) Authentication

Source: Lecture_W6L64_Lecture_26__Cloud_Computing__Security_I | Difficulty: Easy

Q8. What is the primary motivation behind Fog Computing?

- A) To replace cloud computing entirely.
- B) To reduce latency and network load for edge-heavy applications.
- C) To centralize all data processing in one location.
- D) To eliminate the need for wireless networks.

Source: Lecture_W7L75_Lecture_33__Fog_Computing_I | Difficulty: Easy

Q9. What is the main advantage of Docker containers over traditional Virtual Machines (VMs)?

- A) Containers abstract the hardware layer, while VMs abstract the OS layer.
- B) Containers require a full guest operating system, while VMs do not.
- C) Containers share the host OS kernel, making them lighter and faster.
- D) VMs are more portable across different cloud platforms than containers.

Source: Lecture_W8L82_Lecture_36__Introduction_to_DOCKER_Container | Difficulty: Easy

Q10. What is the role of an IoT Gateway in an IoT Cloud architecture?

- A) To perform all data analytics in the cloud.
- B) To act as a central server for all IoT devices globally.
- C) To bridge diverse IoT device protocols to a standard format for the cloud.
- D) To replace the need for cloud storage.

Source: Lecture_W8L85_Lecture_39__IoT_Cloud | Difficulty: Easy

Q11. Which computing paradigm introduced the concept of treating computing as a metered service, where users pay only for what they consume?

- A) Distributed Systems
- B) Grid Computing
- C) Cluster Computing
- D) Utility Computing

Source: Lecture_W1L20_Lecture_2__Cloud_Computing__Overview__contd__ | Difficulty: Medium

Q12. Which scaling method is considered the primary approach for achieving elasticity in cloud computing due to its resilience and cost-effectiveness at large scales?

- A) Vertical Scaling (Scale-Up)
- B) Horizontal Scaling (Scale-Out)
- C) Diagonal Scaling
- D) Static Scaling

Source: Lecture_W1L22_Lecture_4__Cloud_Computing_Architecture | Difficulty: Medium

Q13. In the three-tier IaaS management architecture, which component is responsible for directly interacting with the hypervisor to execute commands like `start` or `stop` a VM?

- A) Cloud Manager
- B) Cluster Manager
- C) Computer Manager
- D) Data Object Storage

Source: Lecture_W2L29_Lecture_7__Cloud_Computing__Virtualization | Difficulty: Medium

Q14. For processing very large XML files or streaming data with minimal memory footprint, which type of XML parser is most suitable?

- A) DOM Parser
- B) SAX Parser
- C) XSLT Processor
- D) XPath Parser

Source: Lecture_W2L31_Lecture_9__Cloud_Computing__XML_Basics_II | Difficulty: Medium

Q15. If a cloud service provider guarantees 99.95% availability and the actual availability falls to 97.38%, which of the following is a likely consequence based on typical SLA penalty tiers?

- A) The provider receives a bonus for good service.
- B) The consumer receives a 10% service credit.
- C) The consumer receives a 25% service credit.
- D) No penalty, as 97.38% is still high.

Source: Lecture_W3L37_Lecture_11__Service_Level_Agreement | Difficulty: Medium

Q16. Which distributed database system, developed by Amazon, prioritizes availability and partition tolerance, often at the cost of immediate consistency, making it suitable for 'always-on' services?

- A) Google File System (GFS)
- B) Bigtable
- C) Dynamo
- D) Hadoop Distributed File System (HDFS)

Source: Lecture_W3L39_Lecture_13__Managing_Data | Difficulty: Medium

Q17. In the MapReduce paradigm, what is the primary role of the 'Reducer' task?

- A) To split the input data into smaller chunks.
- B) To apply a user-defined function to each data chunk and emit key-value pairs.
- C) To aggregate intermediate key-value pairs from mappers and produce a final result.
- D) To manage fault tolerance and task scheduling.

Source: Lecture_W3L40_Lecture_14__Introduction_to_Map_Reduce | Difficulty: Medium

Q18. In Microsoft Azure's architectural blueprint, which computational role is designed for running background processing and distributed tasks without direct user interaction?

- A) Web Role
- B) Worker Role
- C) App Fabric
- D) SQL Azure

Source: Lecture_W4L47_Lecture_17__Cloud_Computing__Case_Study_with_a_Co | Difficulty: Medium

Q19. A server running at 0% load still consumes a significant amount of base power. This observation is a key motivation for which energy-saving strategy in cloud data centers?

- A) Live Migration
- B) VM Image Minimization
- C) Workload Consolidation
- D) Green Cloud Framework

Source: Lecture_W5L58_Lecture_24__Resource_Management_I | Difficulty: Medium

Q20. Which of Gartner's 7 Cloud Computing Security Risks is directly related to the difficulty of proving compliance with regulations when data is held by a third party, possibly in another country?

- A) Privileged User Access
- B) Regulatory Compliance & Audit
- C) Data Location
- D) Data Segregation

Q21. What is the primary mechanism of Mobile Cloud Computing (MCC) that allows resource-constrained mobile devices to overcome their limitations?

- A) Local caching of all data.
- B) Computation Offloading.
- C) Enhanced on-device processing power.
- D) Direct connection to satellite internet.

Source: Lecture_W7L73_Lecture_31__Mobile_Cloud_Computing__I | Difficulty: Medium

Q22. In Sensor Cloud, what is a 'Derived Sensor'?

- A) A physical sensor that has been modified.
- B) A software-based emulation of a physical sensor.
- C) A virtual sensor that generates output by combining data from multiple, often heterogeneous, physical sensors.
- D) A sensor that only provides raw, unprocessed data.

Source: Lecture_W8L84_Lecture_38__Sensor_Cloud_Computing | Difficulty: Medium

Q23. In the context of cloud service models, which statement accurately describes the shared responsibility for the network stack in IaaS, PaaS, and SaaS?

- A) In IaaS, the provider manages the entire network stack; in PaaS and SaaS, the consumer manages it.
- B) In SaaS, the provider manages the entire stack; in PaaS, responsibility extends higher than IaaS, but the consumer still manages applications and data.
- C) In PaaS, the consumer manages the physical network; in IaaS, the provider manages everything.
- D) In all three models, the consumer is responsible for network configuration and firewalls.

Source: Lecture_W1L23_Lecture_5__Cloud_Computing_Architecture__contd__ | Difficulty: Hard

Q24. According to Cloudeconomics, the cloud is more economical than owning infrastructure when which condition is met, where P is peak demand, A is average demand, and U is utility premium?

- A) $P / A < U$
- B) $P / A > U$
- C) $P * A = U$
- D) $P + A = U$

Source: Lecture_W3L38_Lecture_12__Cloud_Economics | Difficulty: Hard

Q25. An in-house server costs \$600,000 over 3 years, with raw hourly cost of \$22.83. If it's only 40% efficient, and a cloud alternative costs \$42/hr (80% efficient) plus \$1/hr management, what is the total cost per effective hour for the in-house server when also including \$22/hr for power and \$6/hr for management?

- A) \$53.50/hr
- B) \$57.08/hr
- C) \$85.08/hr
- D) \$60.61/hr

Source: Lecture_W5L56_Lecture_22__Cloud_Computing__Economics_Tutorial | Difficulty: Hard

Q26. In the 'Hey, You, Get Off of My Cloud' case study, what was the significance of identifying the Dom0 IP address when attempting to verify co-residence?

- A) It indicated the public IP address of the target VM.
- B) A matching Dom0 IP for two instances strongly suggested they were on the same physical machine.
- C) It was used to launch a side-channel attack directly.
- D) It provided the internal IP range of the entire cloud provider.

Source: Lecture_W6L66_Lecture_28__Cloud_Computing__Security_III | Difficulty: Hard

Q27. In collaborative SaaS clouds, which security principle can be inadvertently violated if a user leverages permissions from multiple domains to bypass local constraints, potentially allowing a single user to complete a critical task alone?

- A) Inter-Domain Role Mapping (IDRM)
- B) Risk-Based Access Control
- C) Separation of Duty (SoD)
- D) Cyclic Inheritance Conflict

Source: Lecture_W6L67_Lecture_29__Cloud_Computing__Security_Issues_In_Co | Difficulty: Hard

Q28. An intelligent cloud broker uses a fuzzy inference engine to make decisions. In the fuzzy inference workflow, what is the purpose of 'Fuzzification'?

- A) To convert linguistic variables into crisp numerical values.
- B) To apply a predefined rule base to the inputs.
- C) To map crisp inputs (e.g., 99.98% uptime) to linguistic fuzzy sets (e.g., 'Excellent' availability).
- D) To aggregate the results of all rules into a single suitability score.

Source: Lecture_W6L68_Lecture_30__Cloud_Computing__Broker_for_Cloud_Mark | Difficulty: Hard

Q29. According to the mathematical view presented, offloading a mobile application task to the cloud is most beneficial when:

- A) The task is light on computation and heavy on data transfer.
- B) The network bandwidth is low, and the cloud server is slow.
- C) The task is compute-intensive and light on data transfer, and the network bandwidth is high.
- D) The mobile device has unlimited battery life.

Source: Lecture_W7L74_Lecture_32__Mobile_Cloud_Computing__II | Difficulty: Hard

Q30. In a PHIL experiment for DFIG fault analysis, what is the primary reason asymmetrical faults are considered more detrimental than symmetrical faults of the same voltage dip magnitude?

- A) Asymmetrical faults only affect one phase, making them harder to detect.
- B) They introduce negative sequence components into the system voltage, causing severe and complex disturbances in rotating machines.
- C) Symmetrical faults are always completely mitigated by grid protection.
- D) Asymmetrical faults cause the DFIG to disconnect from the grid immediately.

Source: Lecture_W12L119_Lecture_57__CPS_and_Cloud_Computing | Difficulty: Hard

Answer Key

1: B	2: D	3: B	4: B	5: C
6: C	7: B	8: B	9: C	10: C
11: D	12: B	13: C	14: B	15: C
16: C	17: C	18: B	19: C	20: B
21: B	22: C	23: B	24: B	25: C
26: B	27: C	28: C	29: C	30: B